

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Area and volume

02

10 answers · Rationale-rich · Teach from doc

Q1**66**

The correct answer is 66. It's given that the right circular cone has a height of 22 centimeters cm and a base with a diameter of 6 cm. Since the diameter of the base of the cone is 6 cm, the radius of the base is 3 cm. The volume V , in cm^3 , of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, in cm, and r is the radius, in cm, of the base of the cone. Substituting 22 for h and 3 for r in this formula yields $V = \frac{1}{3}\pi 3^2 22$, or $V = 66\pi$. Therefore, the volume of the cone is $66\pi \text{ cm}^3$. It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is 66.

Q2

D

D. 1,944

Choice D is correct. The area of a rectangle is given by bh , where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle ABCD, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy . Therefore, the area, in square units, of ABCD can be represented by the expression xy . It's given that the length of each side of EFGH is 6 times the length of the corresponding side of ABCD. Therefore, the length, in units, of the base of EFGH can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $6x6y$, which is equivalent to $36xy$. Therefore, the area, in square units, of EFGH can be represented by the expression $36xy$. It's given that the area of ABCD is 54 square units. Since xy represents the area, in square units, of ABCD, substituting 54 for xy in the expression $36xy$ yields 3654, or 1,944. Therefore, the area, in square units, of EFGH is 1,944.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not 6, times the length of the corresponding side of ABCD.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{2}{3}}$, not 6, times the length of the corresponding side of ABCD.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not 6, times the length of the corresponding side of ABCD.

Q3**C**

C. 44

Choice C is correct. The volume of right circular cylinder A is given by the expression $\pi r^2 h$, where r is the radius of its circular base and h is its height. The volume of a cylinder with twice the radius and half the height of cylinder A is given by $\pi(2r)^2\left(\frac{1}{2}\right)h$, which is equivalent to $4\pi r^2\left(\frac{1}{2}\right)h = 2\pi r^2 h$. Therefore, the volume is twice the volume of cylinder A, or $2 \times 22 = 44$.

Choice A is incorrect and likely results from not multiplying the radius of cylinder A by 2. Choice B is incorrect and likely results from not squaring the 2 in $2r$ when applying the volume formula. Choice D is incorrect and likely results from a conceptual error.

Q4**12**

The correct answer is 12. The volume, V , of a right square prism can be calculated using the formula $V = s^2 h$, where s represents the length of an edge of the base and h represents the height of the prism. It's given that the volume of the prism is 2,016 cubic units and the height is 14 units. Substituting 2,016 for V and 14 for h in the formula $V = s^2 h$ yields $2,016 = s^2 14$. Dividing both sides of this equation by 14 yields $144 = s^2$. Taking the square root of both sides of this equation yields two solutions: $-12 = s$ and $12 = s$. The length can't be negative, so $12 = s$. Therefore, the length, in units, of an edge of the base is 12.

Q5**A****A.** 149,796

Choice A is correct. The volume of a cube can be found by using the formula $V = s^3$, where V is the volume and s is the edge length of the cube. Therefore, the volume of the given cube is $V = 68^3$, or 314,432 cubic inches. The volume of a sphere can be found by using the formula $V = \frac{4}{3}\pi r^3$, where V is the volume and r is the radius of the sphere. Therefore, the volume of the given sphere is $V = \frac{4}{3}\pi 34^3$, or approximately 164,636 cubic inches. The volume of the space in the cube not taken up by the sphere is the difference between the volume of the cube and volume of the sphere. Subtracting the approximate volume of the sphere from the volume of the cube gives $314,432 - 164,636 = 149,796$ cubic inches.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**2216**

The correct answer is 2,216. The surface area of a prism is the sum of the areas of all its faces. A right rectangular prism consists of six rectangular faces, where opposite faces are congruent. It's given that this prism has a length of 28 cm, a width of 15 cm, and a height of 16 cm. Thus, for this prism, there are two faces with area $28 \times 15 = 420$ cm², two faces with area $28 \times 16 = 448$ cm², and two faces with area $15 \times 16 = 240$ cm². Therefore, the surface area, in cm², of the right rectangular prism is $2(420) + 2(448) + 2(240)$, or 2,216.

Q7**A****A. 6**

Choice A is correct. It's given that the length of each side of a scale model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of a ballroom. Therefore, the area of the scale model is $\frac{1}{10}^2$, or $\frac{1}{100}$, times the area of the actual floor of the ballroom. It's given that the area of the floor of the ballroom is 600 square meters. Therefore, the area, in square meters, of the scale model is $\frac{1}{100}600$, or 6.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**C**

C. $R = 7r$ and $H = 8h$

Choice C is correct. The volume of a right circular cylinder is equal to $\pi a^2 b$, where a is the radius of a base of the cylinder and b is the height of the cylinder. It's given that the cylinder shown has a radius of r and a height of h . It follows that the volume of the cylinder shown is equal to $\pi r^2 h$. It's given that the second right circular cylinder has a radius of R and a height of H . It follows that the volume of the second cylinder is equal to $\pi R^2 H$. Choice C gives $R = 7r$ and $H = 8h$. Substituting $7r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi 7r^2 8h$, or $\pi 49r^2 8h$, which is equivalent to $\pi 392r^2 h$, or $392\pi r^2 h$. This expression is equal to 392 times the volume of the cylinder shown, $\pi r^2 h$. Therefore, $R = 7r$ and $H = 8h$ could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder.

Choice A is incorrect. Substituting $8r$ for R and $7h$ for H in the expression that represents the volume of the second cylinder yields $\pi 8r^2 7h$, or $\pi 64r^2 7h$, which is equivalent to $\pi 448r^2 h$, or $448\pi r^2 h$. This expression is equal to 448, not 392, times the volume of the cylinder shown.

Choice B is incorrect. Substituting $8r$ for R and $49h$ for H in the expression that represents the volume of the second cylinder yields $\pi 8r^2 49h$, or $\pi 64r^2 49h$, which is equivalent to $\pi 3,136r^2 h$, or $3,136\pi r^2 h$. This expression is equal to 3,136, not 392, times the volume of the cylinder shown.

Choice D is incorrect. Substituting $49r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi 49r^2 8h$, or $\pi 2,401r^2 8h$, which is equivalent to $\pi 19,208r^2 h$, or $19,208\pi r^2 h$. This expression is equal to 19,208, not 392, times the volume of the cylinder shown.

Q9**D**

D. $V = \pi r^3 + 4\pi r^2$

Choice D is correct. The volume, V , of a right cylinder is given by the formula $V = \pi r^2 h$, where r represents the radius of the base of the cylinder and h represents the height. Since the height is 4 inches longer than the radius, the expression $r + 4$ represents the height of each cylindrical container. It follows that the volume of each container is represented by the equation $V = \pi r^2(r + 4)$. Distributing the expression πr^2 into each term in the parentheses yields $V = \pi r^3 + 4\pi r^2$.

Choice A is incorrect and may result from representing the height as $4r$ instead of $r + 4$. Choice B is incorrect and may result from representing the height as $2r$ instead of $r + 4$. Choice C is incorrect and may result from representing the volume of a right cylinder as $V = \pi r h$ instead of $V = \pi r^2 h$.

Q10**8**

The correct answer is 8. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point $(6, 6)$ is halfway between the points $(6, 2)$ and $(6, 10)$, and is a distance of 4 units from each of those two points. The point $(6, 6)$ is also a distance of 4 units from $(2, 6)$. Because the point $(6, 6)$ is the same distance from all three points shown, it must be the center of the circle. Since that distance is 4, it follows that the radius of the circle is 4. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the given circle is $2\pi(4)$, or 8π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is 8.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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